NAME Alejandro Gutierrez

ChE 210 Exam #1 11 October 2019

(1) (25 pts.) Consider a mixture of gasses consisting of three species, namely, A, B, and C. It is known that the mixture is made up of 40 mol% A, 18.75 wt % of B, 20 mol% of C. The molecular weight of A is 40kg/kmole and of C is 50 kg.kmole.

(a) Calculate the molecular weight of B in kg/kmol.

	A	B	C
x	40%	40%	20%
ω	50%	18.75%	31.25%

For 1 kmol basis

$0.4 \text{ kmol } A \times \frac{40 \text{ kg}}{1 \text{ kmol}} = 16 \text{ kg } A$

$0.4 \text{ kmol } B \times \frac{\quad}{\quad} = x \text{ kg } B$

$0.2 \text{ kmol } C \times \frac{50 \text{ kg}}{1 \text{ kmol}} = 10 \text{ kg } C$

$$\frac{x}{16+10+x} = 0.1875$$

$$\frac{x}{26+x} = 0.1875 \Rightarrow x = 0.1875(26+x) \Rightarrow x = 4.875 + 0.1875x$$

$$0.8125x = 4.875$$

$$x = \frac{6 \text{ kg } B}{0.4 \text{ kmol } B} = 15 \frac{\text{kg}}{\text{kmol}} B$$

1	20
2	15
3	24
T	59

$M_B =$ 15 kg/kmol

(b) Calculate the mole average molecular weight of the mixture kg/kmole.

use mol frac

$$\frac{40 \text{ kg}}{\text{kmol}}(0.5) + \frac{15 \text{ kg}}{\text{kmol}}(0.1875) + \frac{50 \text{ kg}}{\text{kmol}}(0.3125)$$

$$= 38.44 \frac{\text{kg}}{\text{kmol}}$$

−5

20

$\bar{M} =$ 38.44 kg/kmol

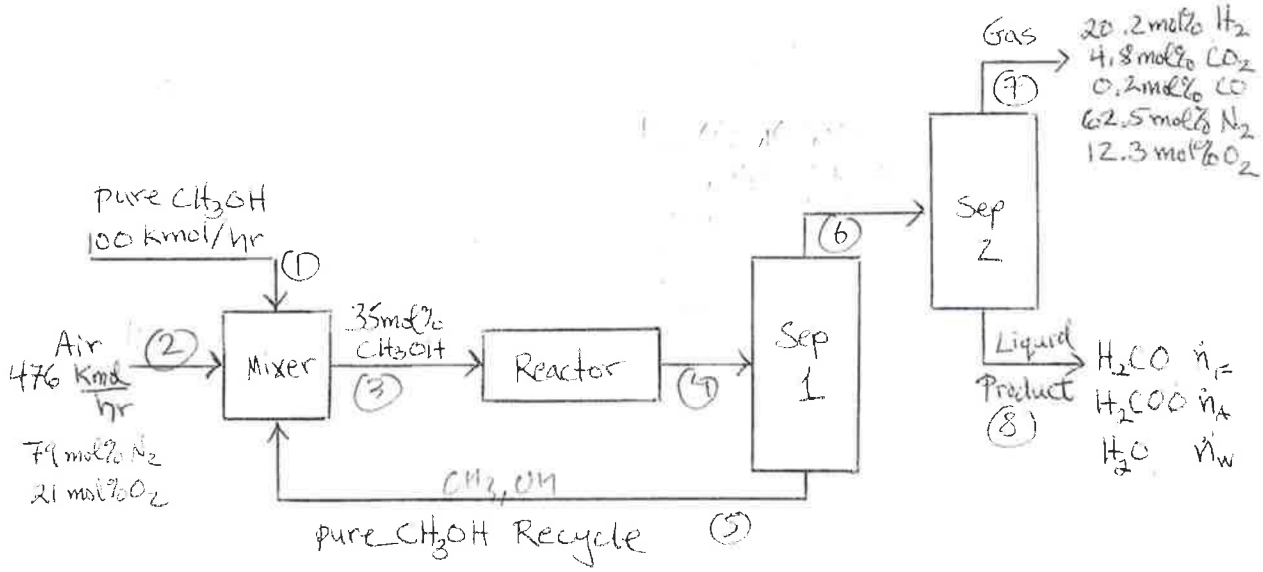
(2) (50 pts.) Formaldehyde (H_2CO) is produced by oxidation of methanol (CH_3OH). As shown, four reactions take place in the reactor. Pure methanol (CH_3OH) at 100 kmol/hr along with 476 kmol/hr of air (21 mol% O_2 and 79 mol% N_2) and a recycle stream of pure methanol are mixed and then enter the reactor. The reactor inlet stream is 35 mol% CH_3OH. The reactor exit stream goes to a separator with one exit stream of pure methanol that is recycled to the mixer and another stream that goes to a second separator that has an exit stream of gas with 20.2 mol% H_2, 4.8 mol % CO_2, 0.2 mol % CO, 62.5 mol% N_2, and 12.3 mol % O_2 and a liquid product stream containing formaldehyde (H_2CO), formic acid (H_2COO), and water (H_2O). See the block flow diagram below.

$$CH_3OH + \frac{1}{2}O_2 \rightarrow HCHO + H_2O \qquad (R1)$$

$$CH_3OH + O_2 \rightarrow HCOOH + H_2O \qquad (R2)$$

$$CH_3OH + O_2 \rightarrow CO + 2H_2O \qquad (R3)$$

$$CH_3OH + \frac{3}{2}O_2 \rightarrow CO_2 + 2H_2O \qquad (R4)$$



(a) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Fill in the table below giving a brief explanation for your entries. Compute the DOF. (10 pts.)

	Number	Explanation
4a. Stream Variables	32	Stream 1 = 1, Stream 2 = 2, Stream 3 = 3, Stream 4 = 9, Stream 5 = 1, Stream 6 = 8, Stream 7 = 5, Stream 8: 3
4b. System Variable	4	4 reactions take place
Total Independent Variables	36	
5a. Specified Flows	2	The mixer has two specified flows
5b. Specified Stream Compositions	5 +1	1 from stream 3, 4 from stream 7 Air is counted as a single component
5c. Specified System Performances	0	
5d. Material Balance Equations	29	For mixer: 3 For S_1: 9 For S_2: 8 For reaction: 9
Total Independent Equations	36	
DOF	0	

very close

10

NAME Alejandro Gutierrez

PROBLEM 2 CONTINUED

(b) Calculate the product stream molar flow rates in kmol/hr for the formaldehyde (H_2CO) $\dot{n}_F$, formic acid (H_2COO) $\dot{n}_A$, and water (H_2O) $\dot{n}_W$ in kmol/hr. *Hint: Consider an atomic balance.* (30 pts.)

Atomic balance

C:

1

$\dot{n}_F$ = ________ kmol H_2CO /min, $\dot{n}_A$ = ________ kmol H_2COO /min, $\dot{n}_W$ = ________ kmol H_2O/min

(c) Calculate the fractional conversion of methanol for the reactor (the single-pass conversion). (10 pts.)

$$f_{methanol} = \frac{\text{mol methanol cons}}{\text{mol fed to reactor}}$$

assume methanol cons = 75 mol

assume mol fed = 100 + 25

4

$$f_{methanol} = \frac{75}{125} = 0.6$$

15

$f_{C,methanol}$ = 0.6 Selectivity = ________